

KUM DONALSIEN AKWO

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PROFESSIONAL PROFILE

Final-year B.Tech Computer Engineering student specialising in Networks and Security at the College of Technology, University of Buea with strong passion for cyber security and cyberspace as a whole. Skilled in Cyber security, network security engineering, AI/ML-based and hybrid intrusion detection systems and other network monitoring and auditing tools. Holds 2 years of hands-on technical experience in network installation, configuration, troubleshooting and security with Telnet Cameroon. Designed and built Bastion IDS, a full hybrid intrusion detection system combining signature-based detection, machine learning ensemble models, deep learning and unsupervised learning models for real-time threat detection. Proven leader as Program Delegate for the Computer Engineering Department, under the Networks and Security option.

EDUCATION

COLLEGE OF TECHNOLOGY, UNIVERSITY OF BUEA

Oct 2023 - Dec 2026

Bachelor of Technology, (B.Tech) Computer Engineering

3.62/4.0

GCE ADVANCED LEVEL

June 2023 Session

General Certificate of Education (GCE) Board, Cameroon

- Biology
- Chemistry
- Pure Mathematics with Mechanics
- Physics
- Computer Science

SKILLS

- Cyber security (Kali Linux, Wireshark, Nmap, Snort, Suricata, IDS/IPS, Ethical Hacking, Penetration Testing, Fortigate, Data Security, Security Operations, Cyber Threat Intelligence, Social Engineering, Information Security)
- Networking (Cisco IOS, Cisco Switches & Routers, OSPF, BGP, IS-IS, RIP, Network Design, Wireless Networking, Network Architecture, IP Addressing)
- Cloud & AWS (AWS EC2, S3, VPC, IAM, CloudFront, CloudWatch, Route 53, VPC design, cloud networking, cloud security)
- Programming (Python, JavaScript, C, Java, PHP, MySQL)
- Frameworks & Tools (FastAPI, React, Arduino, Scikit-learn, XGBoost, CatBoost, Keras/TensorFlow, mitre framework)
- Operating Systems (Windows, Kali Linux)
- Productivity (Microsoft Office Suite)

EXPERIENCE

TELNET CM

August 2024 - September 2024

Intern

- Completed a structured 3-month internship focused on network installation, configuration and security operations
- Gained hands-on experience with physical Cisco switches, routers, Fortigate firewalls and enterprise network infrastructure
- Configured routing protocols (OSPF, BGP, IS-IS, RIP), wireless access points and secure remote access solutions
- Deployed IP-based security surveillance systems and Starlink satellite internet infrastructure on client sites
- Recognised and awarded Best Intern for the batch by TELNET CAMEROON management

TELNET CM

October 2024 - Present

Support Staff

- Provides ongoing technical support and network engineering assistance across client projects
- Assists in planning and executing corporate network infrastructure deployments for enterprise and institutional clients
- Applies cybersecurity monitoring best practices and contributes to network security posture assessments
- Supports installation and configuration of Cisco equipment, Fortigate firewalls and IP surveillance systems

TELNET CM

July 2025 - September 2025

Senior Intern & Intern Instructor

- Returned for a second formal internship in a senior capacity with expanded technical and leadership responsibilities
- Served as lead instructor and mentor for a new batch of junior interns, delivering training in network configuration, troubleshooting and cybersecurity best practices
- Supervised junior intern lab sessions covering Cisco IOS configuration, routing protocols and network security concepts
- Continued developing advanced skills in corporate network deployments and field security implementations

PROJECTS

Bastion IDS, A Hybrid Intrusion Detection System (2025-2026)

Designed and developed a multi-layer hybrid IDS as a capstone project.

Bastion IDS features a four-layer detection mechanism: Layer 1 is the signature engine (46MB ET-Open ruleset, 47,357 signatures), Layer 2 is a machine learning ensemble (Random Forest, XGBoost, CatBoost), Layer 3 is a deep neural network specialist (residual MLP), and Layer 4 is an anomaly detection engine. Developed as a standalone desktop application for real-time network monitoring.

Technologies: *Python, FastAPI, React, Scikit-learn, XGBoost, CatBoost, Keras/TensorFlow, Wireshark, Snort*

Evaluation and Recommendation of Network Monitoring Tools (Aug - Sep 2024)

Evaluated industry-standard IDS/IPS and network monitoring tools to identify and prevent cyber threats. Produced formal analysis and recommendations report to enhance organisational security posture.

Technologies: *Wireshark, Snort, Suricata, Kali Linux, Nmap, Security Onion, Kitsune*

Chat Application and File Transfer System (Dec 2024 - Jan 2025)

Built a real-time multi-client chat application with secure file transfer capabilities using Python socket programming.

Technologies: *Python, Socket Programming*

Traffic Light Control System (Jun - Jul 2025)

Designed and implemented an Arduino-based embedded traffic light control system simulating real-world traffic management logic.

Technologies: *Arduino Uno, C++, Embedded Systems*

LANGUAGE

- English (Fluent)
- French (Basic)

ACTIVITIES

- 237HackFest Capture the Flag (CTF) Competition, organised by Hack The Box Cameroon, gained hands-on experience solving real-world cybersecurity challenges
- Program Delegate, Computer Engineering Department, Option Networks & Security, University of Buea, key liaison between students and university administration

SOFT SKILLS

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|---------------------|---------------------------|-----------------------|
| • Leadership | • Problem Solving | • Time Management |
| • Teamwork | • Effective Communication | • Attention to Detail |
| • Critical Thinking | • Adaptability | • Resilience |

CERTIFICATIONS

- Fortinet Certified Fundamentals (FCF) in Cybersecurity, Fortinet (July 2026, valid to July 2028)
- Cisco Introduction to Cybersecurity, Cisco Networking Academy (September 2025)
- Cybercrime Awareness Certification
- Information Security Certification
- Responsible Data Management
- Communications & Media

ACHIEVEMENTS & AWARDS

- 237HackFest Capture the Flag (CTF) Competition, Hack The Box Cameroon, hands-on experience solving real-world cybersecurity challenges
 - Best Intern Award, Recognised by TELNET CAMEROON management as best intern for the August-September 2024 batch
 - Academic Excellence Recognition, Consistently maintaining a GPA of 3.62/4.0 throughout the degree programme
 - Leadership skills award from the College of Technology for demonstrating outstanding leadership capabilities in school
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